

Culminating Activity

de Pedro, Joenell Conrad Florentino, Ian Miguel

Nacu, Drixyl Reece Ruelan, Ivan Paul Seno, Matthew Anakin

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Contents

Abstract

Provide a concise summary of the study including purpose, methods, results, and conclusions (150–250 words).

Chapter 1: Introduction

Background of the Study

The global demand for software developers continues to rise as organizations accelerate digital transformation and adopt artificial intelligence (AI) in their operations. Industry outlooks show that software development remains a high-growth field, with employment projected to expand much faster than average and with strong median wages that reflect sustained labor demand (U.S. Bureau of Labor Statistics, 2025). This makes it important to examine the factors that shape developer compensation, skill requirements, and career opportunities.

Recent studies also indicate that AI is not only changing how software is built but also reshaping the role of developers. Morgan Stanley Research reported that AI coding tools are expected to enhance productivity and may increase hiring demand as enterprises build more complex applications and address technical debt (Morgan Stanley, 2025). In the same direction, Forbes noted that AI is transforming the software developer role by automating routine coding tasks and shifting developers toward more strategic work such as architecture, integration, and product oversight (Drenik, 2024).

In addition, salary trends suggest that experience and specialized technical skills remain major determinants of developer compensation. Dice reported that software developer salaries continue to increase, with higher pay associated with greater experience and stronger demand for modern technical skills (Dice, 2025). Likewise, digital transformation continues to

require developers to expand their competencies in areas such as cloud computing, AI, machine learning, and DevOps, reinforcing the need to understand how skills and compensation are related in the evolving labor market (RVS Media, 2025).

Given these developments, analyzing the Developer Survey 2026 dataset can provide useful insights into developer skills, salary levels, experience, and perceptions of AI (Stack Overflow, 2026). To systematically examine these relationships, this study will apply a machine learning approach using a multilinear regression model, which enables the analysis of how multiple independent variables, such as years of experience, skill sets, and AI usage, simultaneously influence a dependent variable, particularly developer salary (Suman, 2025; Putra et al., 2026). By leveraging multilinear regression, the study aims to quantify the strength and direction of these relationships, identify significant predictors of compensation, and generate data-driven insights into how evolving technologies and skill demands impact the software development workforce (García et al., 2025; Rodrigues, 2022).

Statement of the Problem

The rapid evolution of technology, particularly the rise of artificial intelligence, has significantly influenced the software development industry. Developers are now expected to continuously adapt by acquiring new skills, while organizations adjust compensation based on experience, expertise, and emerging technological demands. Despite the availability of large-scale survey data, there is still a need to clearly understand how these factors interact and influence developer salaries and roles using quantitative and predictive approaches.

In particular, traditional descriptive analysis alone may not fully capture the combined effect of multiple variables on salary. Thus, there is a need to apply a multilinear regression model to examine how several independent variables, such as skills, experience, and AI impact, simultaneously influence developer compensation.

This study aims to analyze the Developer Survey 2026 dataset to identify key patterns and relationships among skills, experience, salary, and the impact of artificial intelligence. Specifically, it seeks to answer the following research questions:

1. What is the relationship between developers' skills, experience, and salary based on the Developer Survey 2026 dataset using a multilinear regression model?
2. How does the impact of artificial intelligence influence developers' roles and salary levels when analyzed through a multilinear regression approach?

Objectives of the Study

General Objective

To comprehensively investigate the relationships among developer skills, experience, salary, and the impact of artificial intelligence using the Developer Survey 2026 dataset through the application of a multilinear regression model and other data analysis techniques.